1.1 Radian and Degree Measure

Name _____ Period ____ Date _____

Pre-Calculus 1.1 · Student sheet

I can…

- find coterminal, complementary, and supplementary angles.
- convert between degrees and radians.
- find arc length and the area of a sector.

Vocabulary · fill in the missing word

Standard position — an angle with its vertex at the origin and its _____ side on the positive x-axis

Radian — the _____ angle whose arc is as long as the radius

Coterminal angles — angles in _____ position with the same terminal side

Complementary and supplementary angles — _____

Part 1 · How many radii around?

Wrap a radius-length string around the circle.

Radii that fit around the circle	Radii that fit around half the circle	Angle at the center for one radius of arc

Part 2 · Notes and examples (fill in each note, then work the example)

Coterminal angles: add or subtract _____ (or 360°).

Example 1

Find one positive and one negative angle coterminal with $\frac{13\pi}{6}$. In which quadrant is its terminal side?

Example 2

Find the complement and the supplement, if possible.

a. $\frac{2\pi}{5}$

b. $\frac{4\pi}{5}$

c. 2.3

Conversions: 180° = _____ radians. Degrees to radians: multiply by π/180.

Example 3

Convert.

a. 135° to radians

b. −270° to radians

c. $\frac{9\pi}{2}$ to degrees

d. 2 radians to degrees

Arc length: s = _____, with θ in radians.

Example 4

A circle has radius 4 inches. Find the length of the arc cut off by a central angle of 240°.

Area of a sector: A = _____, with θ in radians.

Example 5

A sprinkler sprays water 10 feet and turns through 120°. Find the area it waters.

Part 3 · Partner practice

Partner 1

Convert 210° to radians and $\frac{5\pi}{3}$ to degrees.

Partner 2

Find the arc length on a circle of radius 6 cm with central angle $\frac{5\pi}{6}$.